

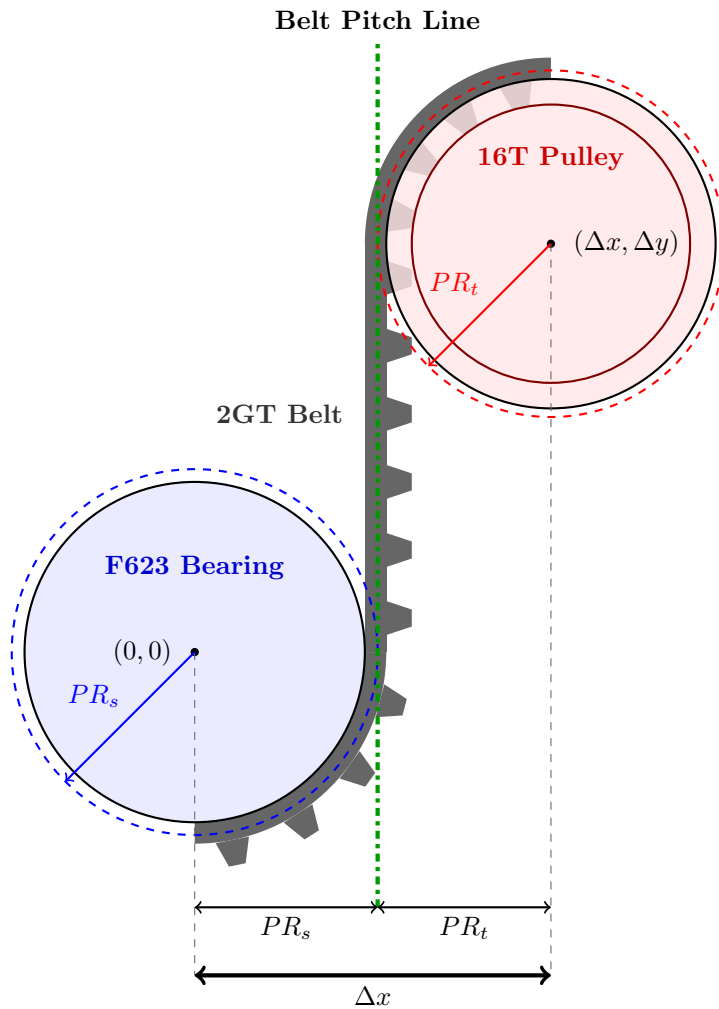
Idler & Pulley Meshing Computation in CoreXY Systems

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1 Introduction

To ensure perfect kinematics in a CoreXY system, the pitch line of the timing belt must remain perfectly parallel to the axis of motion. Here we compute the ideal X-axis offset for a motion configuration pictured below:



1.1 2GT Belt Specifications

- Total Thickness (H): 1.38 mm
- Tooth Height (h): 0.75 mm
- Back/Web Thickness (i): 0.63 mm
- Pitch Line Differential (PLD): 0.254 mm

1.2 Hardware Specifications

- F623 Bearing Outer Diameter: 10.000 mm
- 16T 2GT Pulley Pitch: 2.000 mm
- 16T 2GT Pulley Tooth Count (N): 16

2 Pitch Radius Calculations

2.1 F623 Smooth Idler (PR_s)

When the flat back of the belt rides against a smooth idler, the pitch line is offset outwards. We first calculate the distance from the flat back to the pitch line (t_{pitch}):

$$t_{pitch} = i - PLD = 0.63 \text{ mm} - 0.254 \text{ mm} = 0.376 \text{ mm} \quad (1)$$

Adding this to the radius of the F623 bearing gives the smooth pitch radius:

$$PR_s = \frac{10.000}{2} + 0.376 = 5.376 \text{ mm} \quad (2)$$

2.2 16T Toothed Pulley (PR_t)

For the toothed pulley, the pitch radius is derived strictly from the tooth count and pitch:

$$PR_t = \frac{N \cdot \text{Pitch}}{2\pi} = \frac{16 \cdot 2}{2\pi} \approx 5.093 \text{ mm} \quad (3)$$

3 Ideal X-Offset

Assuming the belt must run perfectly parallel to the Y-axis between the two components, the ideal X-offset is the sum of their respective pitch radii:

$$X_{offset} = PR_s + PR_t = 5.376 + 5.093 = 10.469 \text{ mm} \quad (4)$$